

Spectral Description of a Standard Neural Network

Two layers of a neural network can be interpreted as a directed bipartite graph, as shown in Figure 1.

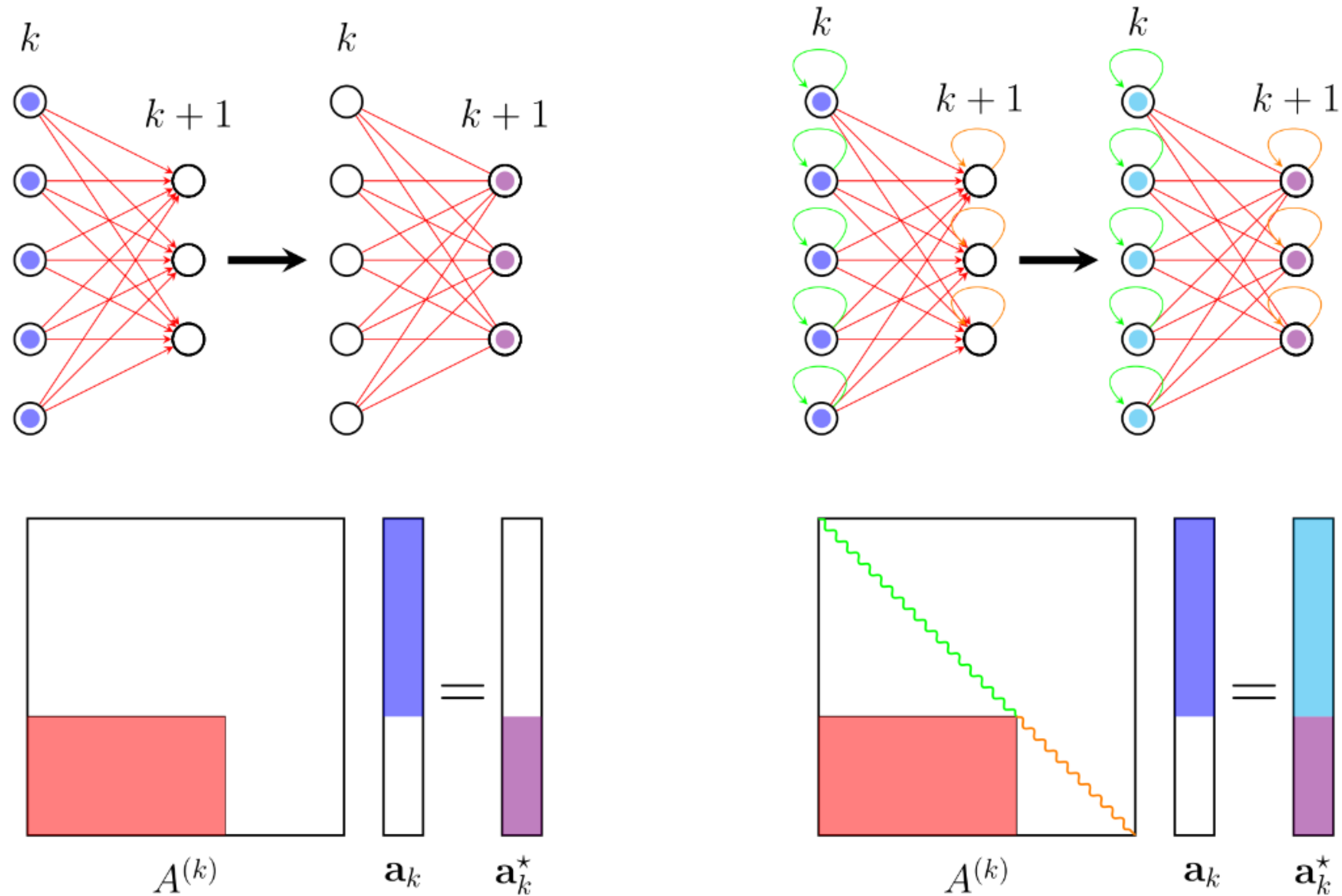


Figure 1. **On the left:** The linear transfer of information between layers k and $k+1$ can be traced back to a directed bipartite graph. **On the right:** representation of the same graph, but with the insertion of *self-connections*; notice that the self-connections do not influence the transfer of information between layers.

Matrix $A = A^{(k)}$ with the inclusion of self-loops (see left panel of Figure 1) can be decomposed as $A = \Phi \Lambda \Phi^{-1}$. The structure of the matrices Φ and Λ is illustrated in Figure 2. We also remark that $\Phi^{-1} = 2\mathbb{I} - \Phi$.

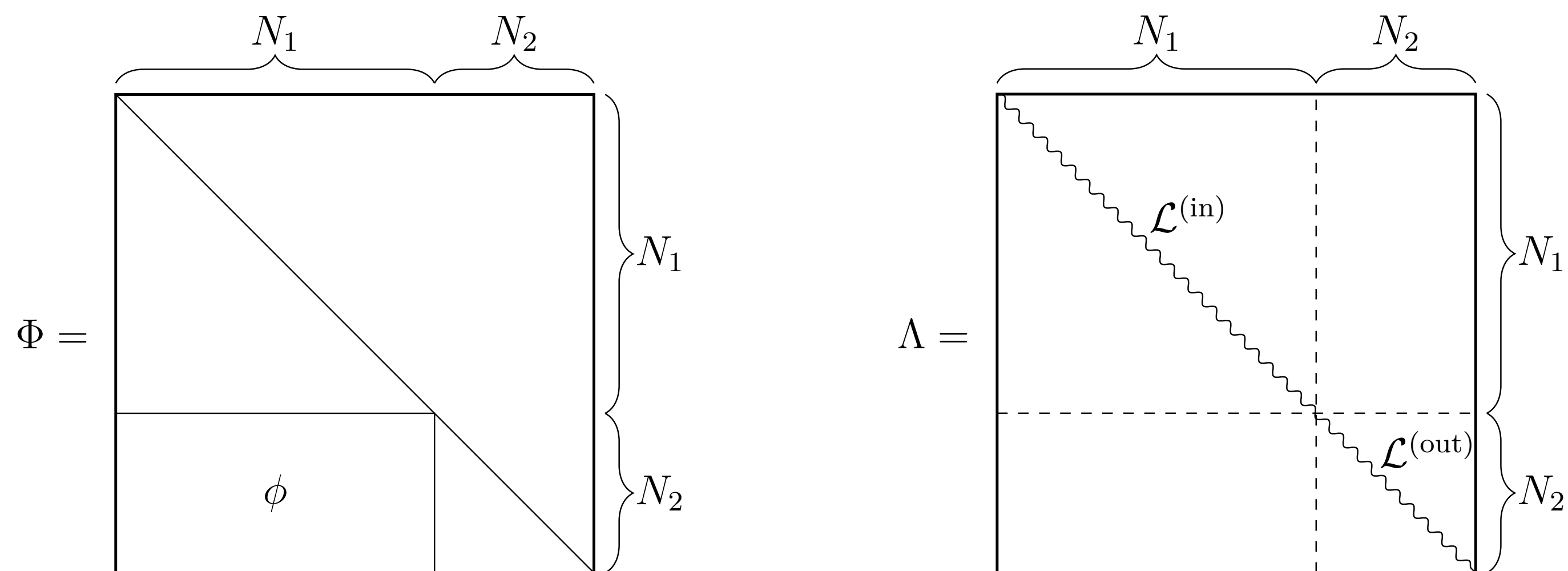


Figure 2. Spectral description of a directed bipartite graph with two layers.

Spectral parametrization of two layers

The weight matrix $\mathcal{W}$, that describes the linear transfer of activities between two adjacent layers, admits the following parametrization in terms of the eigenvalues and eigenvectors of the adjacency matrix:

$$\mathcal{W} = \phi \mathcal{L}^{(in)} - \mathcal{L}^{(out)} \phi \rightarrow w_{ki} = (\lambda_i^{(in)} - \lambda_k^{(out)}) \phi_{ki}. \quad (1)$$

The elements of ϕ , as well as the diagonals of $\mathcal{L}^{(in)}$, $\mathcal{L}^{(out)}$, become the parameters of the model; on which we can perform a gradient descent optimization.

Higher-Order Neural Networks

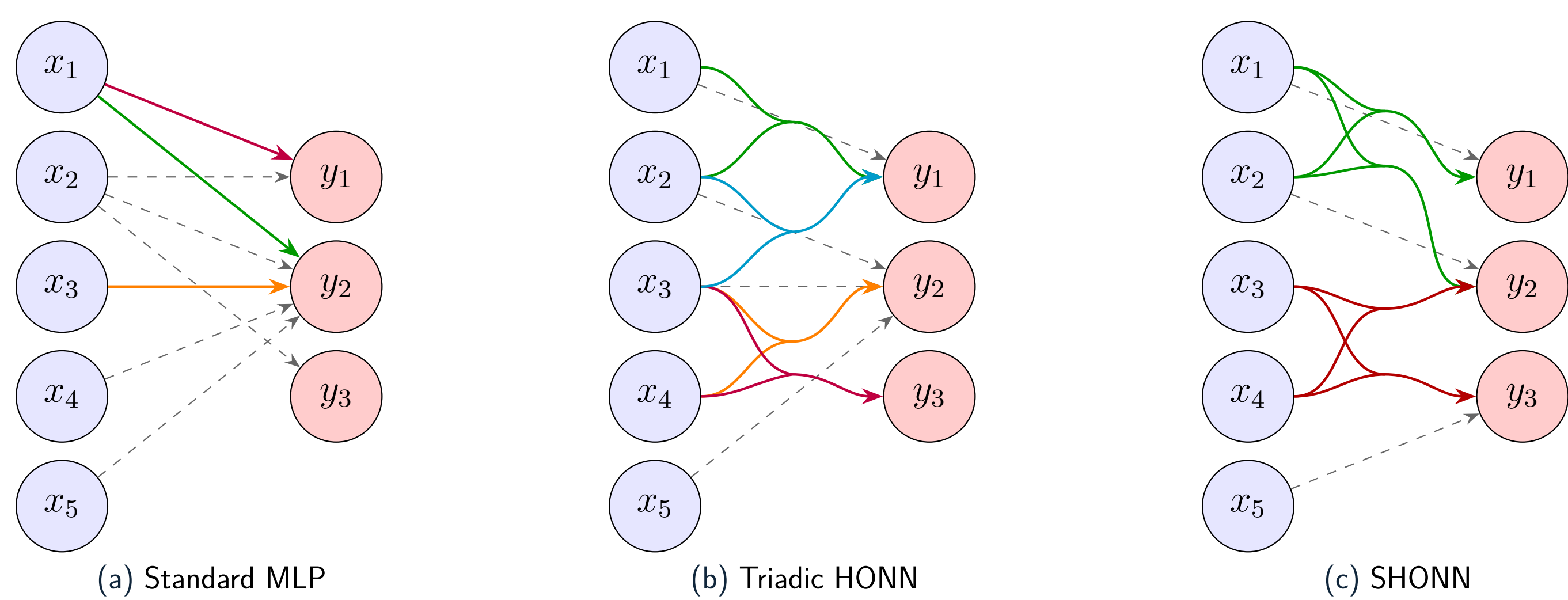


Figure 3

Higher-order neural networks (HONNs) generalize by including higher-order interactions between neurons

$$y_k = \sum_{i=0}^N w_{ki} x_i + \sum_{0 \leq i \leq j \leq \dots \leq n}^{N-1} f(x_i, x_j, \dots, x_n; \theta(k, i, j, \dots, n)); \quad (2)$$

where the simplest member of this family is, of course, the **triadic HONN with product interactions**:

$$y_k = \sum_{i=0}^N w_{ki} x_i + \sum_{0 \leq i \leq j}^{N-1} \tilde{w}_{kij} x_i x_j, \quad (3)$$

Despite their greater expressiveness, HONNs have been historically limited by the $O(N^3)$ parameter scaling of triadic interactions.

We introduce **Spectral Higher-Order Neural Networks (SHONNs)**, a novel paradigm that leverages spectral reparameterization to reduce this complexity to $O(N^2)$:

$$y_k = \sum_i (\lambda_i^{(in)} - \lambda_k^{(out)}) \phi_{ki} x_i + \sum_{i \leq j} (\tilde{\lambda}_{ij}^{(in)} - \tilde{\lambda}_k^{(out)}) \tilde{\phi}_{kij} x_i x_j. \quad (4)$$

This breakthrough allows us to incorporate triadic interactions with $O(N^2)$ parameter scaling by training only the eigenvalues $\tilde{\lambda}_{ij}^{(in)}$, $\tilde{\lambda}_k^{(out)}$ in the triadic portion.

Spectral Parametrization of Triadic Networks

In equation (3) the $\tilde{w}_{kij}$ tensor element can be thought as a $\tilde{w}_{ki'}$ matrix element, with i' being an appropriate index enumerating the set of unordered pairs (i, j) , $0 \leq i \leq j < N$ in lexicographical order:

$$(0, 0) \rightarrow 0, (0, 1) \rightarrow 1, \dots, (0, N-1) \rightarrow N-1, (1, 1) \rightarrow N, \dots, (N-1, N-1) \rightarrow \frac{N(N+1)}{2} - 1.$$

In closed form, i' is given by:

$$i' = j - i + \sum_{k=0}^{i-1} (N - k). \quad (5)$$

The lexicographical bijection allows the deployment of the spectral parametrization for triadic interactions: we first map the tridimensional weight tensor $[\tilde{w}_{kij}]$ into the weight matrix $[\tilde{w}_{ki'}]$, and then express $[\tilde{w}_{ki'}]$ in spectral form. From (3), we get:

$$y_k = \sum_i (\lambda_i^{(in)} - \lambda_k^{(out)}) \phi_{ki} x_i + \sum_{i'} (\tilde{\lambda}_{i'}^{(in)} - \tilde{\lambda}_k^{(out)}) \tilde{\phi}_{ki'} x_{i'}, \quad (6)$$

with $x_{i'} = x_i x_j$. Finally, expanding i' back to two dimensions we get equation (4).

Perceptron Performances: MNIST and Fashion-MNIST

To validate the basic functionality of the proposed higher-order paradigm, we tested perceptron architectures on MNIST and FASHION-MNIST. The results show clear performance and stability gaps in favor of the spectral triadic models. In particular, the direct space triadic networks seem to suffer from confidence saturation problems.

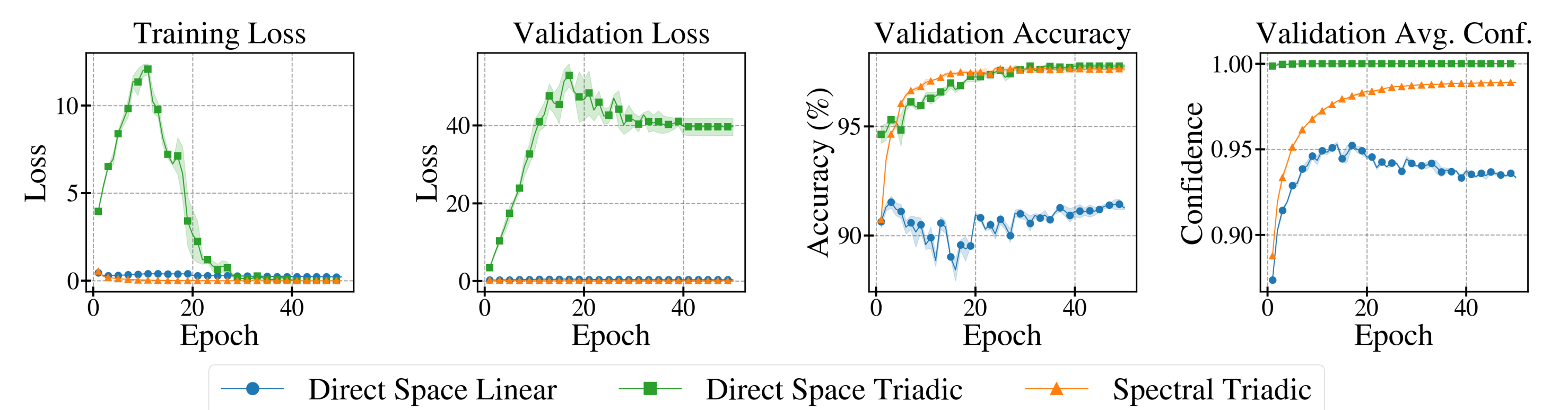


Figure 4. Results of perceptron training on MNIST.

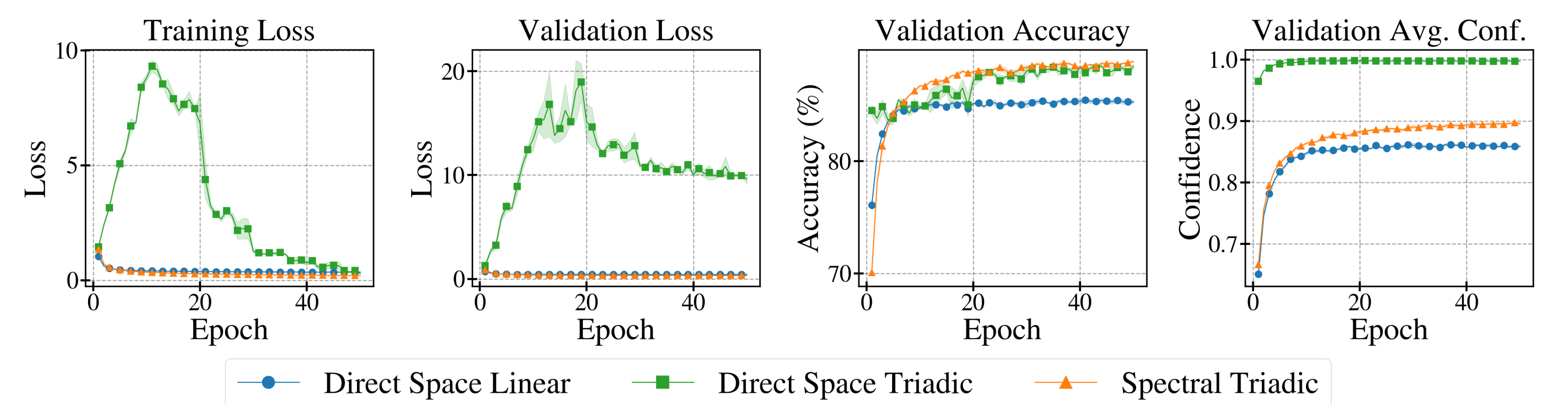


Figure 5. Results on FASHION-MNIST, confirming the same trend observed on MNIST.

Universal Approximation Without Activation

For spectrally parametrized, triadic eigenvalues only, HONNs the following theorem holds.

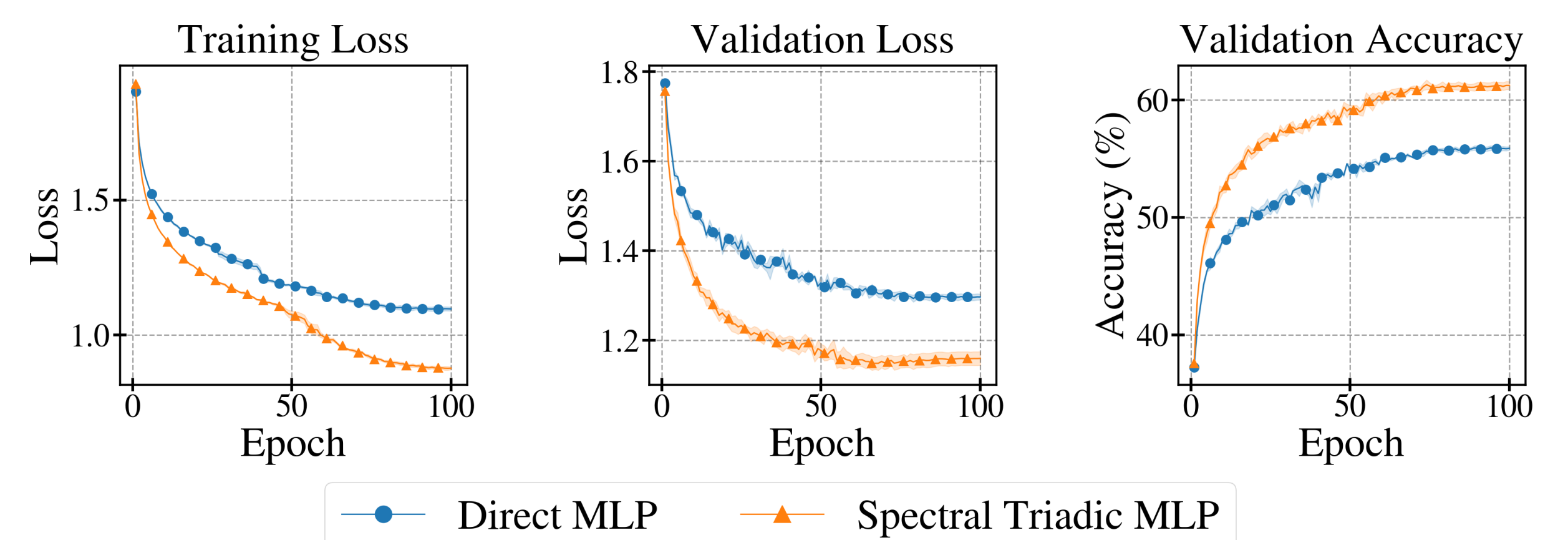
Theorem: Universal Approximation

The space of functions defined by iteratively composing HONN layers with spectral parametrization is dense in $C(X, \mathbb{R}^n)$ on the compact set X .

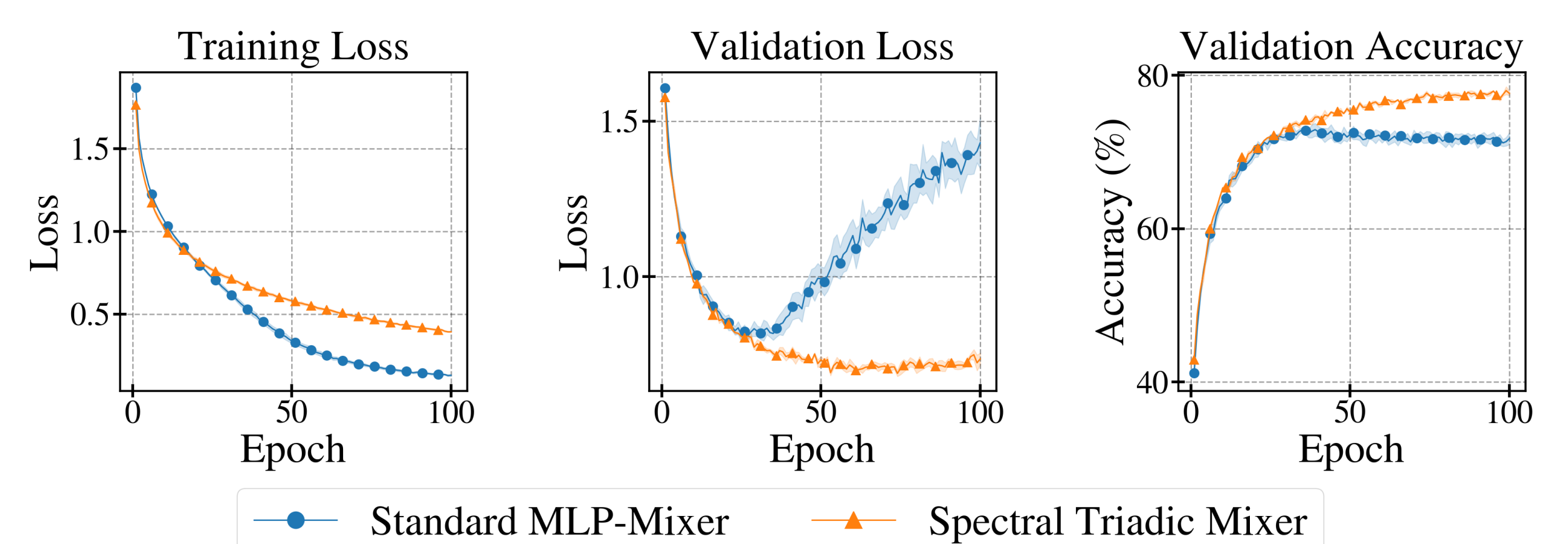
Crucially, this theorem holds **without explicit, element-wise activation functions** (like ReLU or Tanh). The non-linearity is baked directly into the higher-order topology $(x_i x_j)$. By stacking layers, we generate polynomials of degree 2^{N-1} , capable of approximating any continuous function.

Deep Architectures Performances: CIFAR-10

We tested deep architectures on CIFAR-10.



(a) Standard vs Triadic MLPs (4 layers). The triadic model shows a significant performance boost.



(b) Standard vs Triadic MLP-Mixers. In the triadic model the MLP blocks have been replaced with triadic perceptrons. Once again the triadic model outperforms the standard one, but this time seemingly by acting as an implicit regularizer.

Figure 6